

Interactive Color Manipulation with Natural Language Modifiers

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CCS CONCEPTS

• **Applied computing** → **Media arts**; • **Human-centered computing** → **Natural language interfaces**.

KEYWORDS

Color tools, digital painting, language-based editing

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1 INTRODUCTION

Selecting colors is crucial for creating digital illustrations. However, achieving the desired colors, especially for inexperienced users, is difficult because the process involves unintuitive tuning of multiple parameters. In this paper, we propose an interactive color manipulation method allowing users to explore a color space using color modifiers based on natural language expressions such as “lighter”. With our method, users can modify colors using word-based modifier sliders. First, they select a modifier word, and the system then generates a modifier-based slider using a machine learning model. Additionally, our method allows users to edit multiple colors simultaneously while preserving the relationships among the original colors. We conducted a user study with amateur artists and novice users. The results indicate that our method can be inspiring, especially for novices, and can be particularly useful when combined with existing tools to support the creation and exploration of colors.

2 METHOD

Our system was implemented as a Web application based on Python 3 and JavaScript using the Flask framework.

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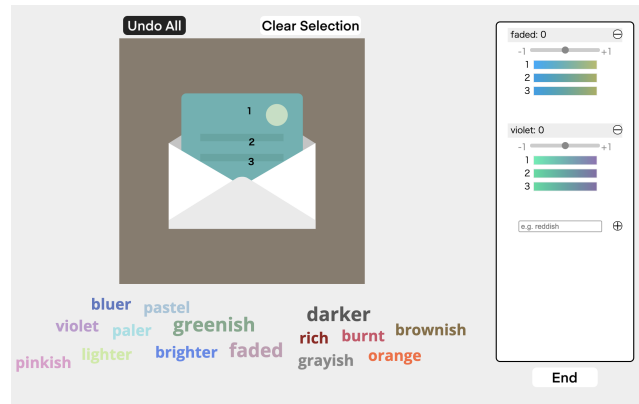


Figure 1: Screenshot of our system. Users input a modifier either by clicking on a word in the modifier word cloud (bottom left) or by typing it into the text box (middle right).

2.1 Color Transformation Model

Han et al. [1] proposed a model that predicts color values $\vec{t} \in \mathbb{R}^3$ from a color modifier such as “lighter” and a reference color $\vec{r} \in \mathbb{R}^3$ in the RGB space. To generalize Winn and Muresan’s model [5] in the RGB color space, they defined their model as follows:

$$\vec{t} = M\vec{r} + \vec{\beta}, \quad (1)$$

where $M \in \mathbb{R}^{3 \times 3}$ is a transformation matrix and $\vec{\beta} \in \mathbb{R}^3$ is a vector.

We trained a network to directly estimate M and $\vec{\beta}$ from modifiers by using a dataset of Winn and Muresan [5]. Note that instead of 300-D Word2Vec embeddings [2, 3] used by Han et al., we adopted a GloVe 100-D representation [4] to keep our application lightweight. Although the network can be applied to any arbitrary word registered in the pre-trained model, this model does not work well for certain words. Then, we chose to select several modifiers that produce satisfactory colors empirically (see Figure 1 (bottom left)).

2.2 Interactive Manipulation

Our color editing process consists of (1) selecting regions in an input image, (2) selecting modifiers, (3) adjusting color values in the selected regions using the sliders associated with the modifiers, and (4) repeating these procedures until the colors are acceptable.

2.2.1 Selecting regions. Users click on a region in an input image and the system selects the clicked region (and clears any previously selected regions). Additionally, users can make multiple selections

Table 1: Seven-point Likert scale questionnaire.

#	Question Items
Q1	I was able to have a wide perspective.
Q2	I was able to discover new colors.
Q3	I was inspired by the options provided by the system.
Q4	I was able to make fine adjustments to the colors.
Q5	I was able to change the colors significantly.

by holding down the shift key while selecting. In contrast, the *Clear Selection* button allows users to clear all selected regions. To improve visibility, the system assigns label IDs to the selected regions (see Figure 1) and flashes the regions once every 1.5 second. This flashing stops temporarily when a user is adjusting a slider.

2.2.2 Slider creation. When users click one color modifier in the word cloud (or type it in a text box), the system displays a modifier slider based on the color transformation model and calculates an adjusted color value $\tilde{c}$ in the RGB space as follows:

$$\tilde{c} = (1 - w)\tilde{r} + w\tilde{t}, \quad (2)$$

where $w \in [-1.0, 1.0]$ is the weight value, $\tilde{r}$ is the color value of the selected region (reference color), and $\tilde{t}$ is a predicted color from the color transformation model.

By adjusting this slider, users can freely change the weight value w . When the weight value is changed, the system calculates the adjusted color values $\tilde{c}$ and colors the selected regions. If a user moves the knob to the right ($= +1.0$), the color moves closer to the modifier. Conversely, if a user moves the knob to the left ($= -1.0$), the color moves away from the modifier. Note that users can delete unnecessary sliders by clicking the minus button in the upper-right corner (see Figure 1).

3 USER STUDY

To evaluate the effectiveness of the proposed modifier sliders, we compared our sliders against RGB/HSV sliders that have the same functionality and are commonly found in traditional color pickers. First, we recruited 16 participants ($= 8$ amateur artists with experience in creating digital illustrations and 8 novice users). Next, we provided them with several SVG images and asked them to recolor the images by using each method. For each method, participants were only presented with a short sentence describing the recoloring target (e.g., “sad in a forest” and “angry in a cave”). At the end, they completed a seven-point Likert scale questionnaire (see Table 1) to evaluate each method. We also conducted a semi-structured interview to gather additional feedback regarding our method and gain a better understanding of user experiences. The entire user study lasted for approximately one hour per participant.

3.1 Results

Table 2 shows the average scores and the results of a Wilcoxon signed-rank test. First, our method achieves significantly higher scores on Q3, suggesting that our method can be more inspirational compared to the baseline method. Second, the baseline method scores significantly higher on Q4, indicating that RGB and HSV sliders are more suitable for making subtle changes. The scores

Table 2: Results of the post-experiment questionnaire. Bold text indicates areas with significant differences.

	Average $\pm$ Std.		p -Value
	Baseline	Ours	
Q1	4.06 $\pm$ 1.83	4.94 $\pm$ 1.51	1.72×10^{-1}
Q2	4.88 $\pm$ 2.00	5.18 $\pm$ 1.41	5.86×10^{-1}
Q3	3.44 $\pm$ 1.41	5.69 $\pm$ 1.36	2.63×10^{-3}
Q4	5.63 $\pm$ 1.28	3.44 $\pm$ 1.93	4.88×10^{-4}
Q5	5.38 $\pm$ 2.49	5.25 $\pm$ 2.39	7.54×10^{-1}

from the amateur artists may have been affected by the fact that they are accustomed to tuning RGB and HSV parameters as part of their regular production processes. In case of the other questions, the differences are not statistically significant.

From the participant comments, we confirmed that modifier sliders can be useful for both creating colors and exploring colors. For example, four participants answered that our method is suitable for novice users: “RGB and HSV parameters are difficult to handle for users who are not familiar with the theory behind them. In contrast, with modifier sliders, it is easy to set the direction of change for such users” and “Modifier sliders can be an alternative for novice users such as young children and elderly people.” In addition, three participants stated that they found the modifier sliders to be enjoyable. One amateur artist said that “for those who are interested in words, it could be a new and fun experience to associate colors with words.” Additionally, another amateur artist appreciated the ability to customize sliders to meet user preferences. In contrast, six participants commented that RGB and HSV sliders are suitable for making subtle changes and it is more useful if the modifier sliders are combined with existing tools such as RGB and HSV sliders, planes, color wheels, color palettes, and eyedroppers.

4 FUTURE WORK

We plan to expand our vocabulary (e.g., “sea” and “heroic”) and offer a wide variety of word choices in different languages. This will not only facilitate editing directions, but also reduce the user effort required to search for a modifier that matches user intentions.

ACKNOWLEDGMENTS

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